

✓ Two sum

Question

Given an integer array `nums` of unique elements, return all possible subsets (the power set).

The solution set must not contain duplicate subsets. Return the solution in any order.

Example 1:

- Input: `nums = [2,7,11,15]`, `target = 9`
 - Output: `[0,1]`
 - Explanation: Because `nums[0] + nums[1] == 9`, we return `[0, 1]`.
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Constraints

- $2 \leq \text{nums.length} \leq 10^4$
 - $-10^9 \leq \text{nums}[i] \leq 10^9$
 - $-10^9 \leq \text{target} \leq 10^9$
 - Only one valid answer exists.
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Solution

so for this question i do it by brute force approach first. Its pretty simple approach, here we check every element of array for i th element and if if we get to our target then we simply return an array of size 2 having value of indices of these elements

Code

```
class Solution {
    public int[] twoSum(int[] nums, int target) {
        int[] result = new int[2];
        Arrays.fill(result, -1);
        for (int i = 0; i <= nums.length - 1; i++) {
            for (int j = i + 1; j <= nums.length - 1; j++) {
                if (nums[i] + nums[j] == target) {
                    result[0] = i;
                    result[1] = j;
                    break;
                }
            }
        }
    }
}
```

```

    }
    }
    return result;
}
}

```

✓ dry run of above code

result -->	<table><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>1</td></tr></table>	-1	-1	0	1	nums -->	<table><tr><td>2</td><td>7</td><td>11</td><td>15</td></tr><tr><td>0</td><td>1</td><td>2</td><td>3</td></tr></table>	2	7	11	15	0	1	2	3
-1	-1														
0	1														
2	7	11	15												
0	1	2	3												
		target -->	26												
ith loop	jth loop	if(num[i] + nums[j] == 26)													
0	1	2+7 = 9 --> false													
	2	2+11 = 13 --> false													
	3	2+15 = 17 --> false													
1	2	7+11 = 18 --> false													
	3	7+15 = 22 --> false													
2	3	11+15 = 26 --> true													
result -->	<table><tr><td>2</td><td>3</td></tr></table>	2	3	→ ANSWER											
2	3														

time complexity = $O(n^2)$

space complexity = $O(1)$

